

Characteristics of an Effective Problem Solver

Computational Design Thinking |
B.Tech 1st Year

What is an Effective Problem Solver?

- An effective problem solver understands the problem clearly.
- Uses logic, creativity and systematic thinking.
- Selects an appropriate solution and checks whether it works.

1. Problem Understanding

- Understand what the actual problem is before solving it.
- Example: If a laptop cannot connect to Wi-Fi, first identify whether the issue is with the laptop, Wi-Fi or router.

2. Analytical Thinking

- Break a complex problem into smaller and manageable parts.
- Example: For a wrong program output, check Input → Algorithm → Code → Output step-by-step.

3. Critical Thinking

- Examine information carefully and judge whether it is correct and useful.
- Example: Before using an online solution, check whether it is correct and suitable for the problem.

4. Creativity

- Think of new and different solutions to a problem.
- Example: To reduce paper use, suggest online assignments, digital notices and e-books.

5. Logical Reasoning

- Use logic, facts and proper reasoning instead of assumptions.
- Example: To find the larger of two numbers, compare A and B and select the larger value using a clear condition.

6. Decision-Making

- Compare available options and select the most suitable one.
- Example: For travelling to college, compare two routes based on distance, traffic and time, then choose the suitable route.

7. Flexibility

- Be ready to change the approach when the current approach does not work.
- Example: If an attendance system has a complex login process, simplify it after receiving user feedback.

8. Persistence

- Continue working on a difficult problem instead of giving up.
- Example: While debugging a program, repeatedly find, fix and test errors until the program works.

9. Communication

- Explain the problem, idea, process and solution clearly to others.
- Example: In a team meeting, explain an online attendance system through Problem → Solution → Working → Result.

10. Evaluation

- Check whether the implemented solution is correct and effective.
- Example: Test a calculator program with different inputs and check whether it produces the expected output.

Quick Revision

- Problem Understanding → Understand the actual problem
- Analytical Thinking → Break the problem into parts
- Critical Thinking → Check and judge information
- Creativity → Generate new ideas
- Logical Reasoning → Use logic and facts
- Decision-Making → Select the suitable option